

Homework #3 — Simple Linear Regression (15 pts)

MATH 240 — Introduction to Statistics, Fall 2026

Due Date: Friday, 9/25 at 11:59pm ET

Instructions

- Please submit this assignment to Gradescope as two separate .pdf files.
 - Answers to all textbook problems as a .pdf file in the HW3: Textbook Problems dropbox on Gradescope.
 - Knitted Quarto document as a .pdf to Palmer Penguins problem in the HW3: Palmer Penguins dropbox on Gradescope.
- Responses can be hand-written or typed for the textbook problems. You will write code with RStudio for Part II.
- Certain questions will ask you to do the problem in a specific way. Please adhere to these instructions or points will be deducted.
- For all questions involving manual calculations, please show all work and round answers to four decimal places.
- For all questions involving code, please provide your code along with the output.
- Refer to syllabus for late work deductions.

Background

We will focus on simple linear regression in this homework, looking at items such as correlation, a least-squares line, residuals, interpretations, and predictions. This will be done with textbook problems and the use of R for the Palmer Penguins dataset.

Note that these concepts will come up throughout the entire week, so the homework will be done as the week goes on.

Part I: Textbook Problems (10 pts)

Complete the following problems:

1. Exercises from Chapter 7: #4, #6, #20, #22, #24

Part II: Palmer Penguins (5 pts)

We will continue using the penguins dataset from the palmerpenguins package.

We will be using R to work with the data. For this question, you will turn in a separate, knitted Quarto document with code and output related to fitting a linear regression model. Follow all instructions below.



Create a new Quarto document. Title the document “HW3, Part II” and add your name in the Name box. Use the following code to bring the penguins dataset into your R environment:

- `library(palmerpenguins)`
- `library(dplyr)`
- `library(ggplot2)`
- `data(penguins)`

Answer the following questions below:

1. (0.25 pts) Filter the penguins dataset using the dplyr package by removing any observations where NA exists in the `flipper_length_mm` variable. Then assign this action to a new object named `penguins_clean`.
 - Hint: `!is.na()` will be a way to determine if a variable has NAs or not.
2. (0.25 pts) Assign objects `x` and `y` to the `penguins_clean$flipper_length_mm` and `penguins_clean$bill_length_mm` variables respectively. We will use `x` and `y` for the following questions.
3. (1 pt) Find the following values using the formulas mentioned in the slides/textbook, utilizing base R functions to get components of the formulas:

- (i) r
 - (ii) $\hat{\beta}_0$
 - (iii) $\hat{\beta}_1$
4. (0.5 pts) Create a scatter plot of flipper length and bill length. In conjunction with the correlation found previously, describe the relationship of these two variables in 2-3 sentences.
 5. (0.5 pts) Fit a best-fit linear regression model to determine bill length using flipper length via the `lm()` function. Store this as an object named `mod`.
 6. (1 pt) Print a summary of the `mod` object. Then,
 - (i) Identify $\hat{\beta}_0$ and $\hat{\beta}_1$. Provide a meaningful, contextual interpretation for each of the two β coefficients.
 - (ii) Determine if flipper length is a significant predictor of bill length.
 - (iii) Find the strength of the fit.
 7. (1 pt) Use the following code:
 - `par(mfrow = c(2, 2))`
 - `plot(mod)`and write a sentence for each plot, describing what it entails and how it impacts our model.
 8. (0.5 pts) With the least-squares line, predict the bill length using the following flipper lengths:
 - 100, 180, 200, 225, 230 mm